

Day 5: Review and Practice Practical

Task 1 :- Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

ANS :-

Here is how to do it quickly:

1.Add Devices :

Drag 1 Wireless Router (WRT300N) and 3 PCs onto the canvas.

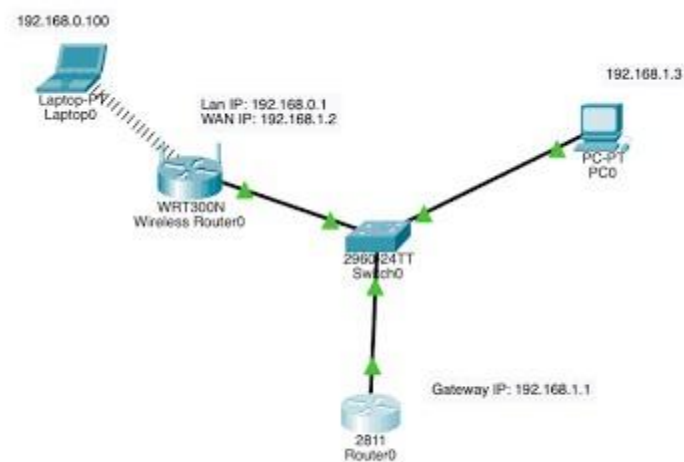
2.Make PCs Wireless :

For each PC: Click it → Turn power OFF → Remove the wired port → Add WMP300N module → Turn power ON.

3.Save File:

Go to File → Save As → Name it HomeWiFiNetwork.pkt.

Check: Dashed wireless lines will connect all 3 PCs to the router automatically.



Task 2 :- Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

ANS :-

1. Router Setup

Set your Router IP to 192.168.10.1 (Subnet: 255.255.255.0).

2. PC Setup

Click each PC → Desktop → IP Configuration → select Static:

- PC 1:
 - IP Address: 192.168.10.2 ◦
Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
- PC 2:
 - IP Address: 192.168.10.3
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
- PC 3:
 - IP Address: 192.168.10.4
 - Subnet Mask: 255.255.255.0 ◦
Default Gateway: 192.168.10.1

Check: Open Command Prompt on PC 1 and type ping 192.168.10.2. If you see "Reply from 192.168.10.2", it works!

Task 3 :- Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

ANS :-

1.Ping via PC Command Prompt : Method 1 :

Command Line Interface.

1. Click on PC 1 → select the Desktop tab → open Command Prompt.
2. Ping the Router (Default Gateway) :

```
ping 192.168.10.1
```

3.Ping PC 2 :

```
ping 192.168.10.3
```

4.Ping PC 3:

```
ping 192.168.10.4
```

2.Ping via Simple PDU Tool (Envelope Icon) : Method 2 : Visual Simulation.

1. Click the Add Simple PDU button on the right toolbox menu (or press P).
2. Click on PC 1 (source device), then click on PC 2 or the Router (destination device).

3. Check the bottom-right Simulation Panel / Realtime Window: look for the status Successful.

Expected Results

When running the ping command in the Command Prompt, you should receive continuous reply messages showing zero packet loss:

```
Pinging 192.168.10.3 with 32 bytes of data:  
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128  
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128  
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128  
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128  
  
Ping statistics for 192.168.10.3:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

How to verify this step was successful: Confirm that all 4 reply lines show Reply from ... with 0% loss. If you see Request timed out, double-check that all devices share the same subnet mask (255.255.255.0) and IP range (192.168.10.x).

Task 4 :- Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

ANS :-

Wireless Network Lab Documentation

Cisco Packet Tracer Architecture, Addressing Scheme & Implementation Guide

Project File: HomeWiFiNetwork.pkt
Topology: Wireless Star (IEEE 802.11n)
Security: WPA2-Personal (AES)

Subnet Mask: 255.255.255.0 (/24)
Gateway IP: 192.168.10.1
Status: **Verified & Operational**

1. Executive Summary & Topology Map

This document detailly records the design, component selection, hardware adaptation, IP addressing scheme, and connectivity verification for the simulated wireless home network standard HomeWiFiNetwork . pkt. The network consists of a central wireless router (Cisco Linksys WRT300N) and three endpoint PCs equipped with WMP300N wireless PCI cards operating on a single broadcast domain.

Logical Topology Screenshot (Packet Tracer Workspace)

2. Complete Network Addressing Table

Device Name	Interface Type	IP Address	Subnet Mask	Default Gateway	SSID / Security
Wireless Router	LAN / Wireless	192.168.10.1	255.255.255.0	N/A (Gateway Node)	HomeNetwork / WPA2-Personal
PC0	Wireless (WMP300N)	192.168.10.2	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal
PC1	Wireless (WMP300N)	192.168.10.3	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal
PC2	Wireless (WMP300N)	192.168.10.4	255.255.255.0	192.168.10.1	HomeNetwork / WPA2-Personal

Packet Tracer Network Lab Documentation

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3. Detailed Step-by-Step Configuration Guide

Step 1: Workspace Assembly & Modular Hardware Setup

1. Drag a **WRT300N Wireless Router** from Network Devices > Wireless Devices into the workspace.
2. Place three standard **PCs (PC0, PC1, PC2)** around the router.
3. By default, Packet Tracer PCs ship with an RJ-45 FastEthernet module (PT-HOST-NM-1CFE). Convert

- Toggle the physical power switch to **OFF**.
- Drag the existing Ethernet module down to the bottom module bin to remove it.
- Insert the **WMP300N 2.4GHz Wireless Module** into the empty PCI slot.
- Toggle the power switch back to **ON**.

Step 2: Router Wireless & LAN Subnet Configuration

1. Click the **WRT300N Router** and navigate to the **GUI** tab.
2. Under **Setup > Basic Setup**:
 - Change IP Address to **192.168.10.1**.
 - Verify Subnet Mask is set to **255.255.255.0**.
 - Click **Save Settings** at the bottom.
3. Navigate to **Wireless > Basic Wireless Settings**:
 - Set Network Name (SSID) to **HomeNetwork**.
4. Navigate to **Wireless > Wireless Security**:
 - Security Mode: **WPA2 Personal** | Encryption: **AES** | Passphrase: **Cisco12345**.

Step 3: Static IP Assignment & Gateway Mapping on Endpoints

Assign unique IP addresses and map the default gateway on each endpoint PC via the **Desktop > IP Configuration** utility:

IP Address:	192.168.10.2
Subnet Mask:	255.255.255.0
Default Gateway:	192.168.10.1
DNS Server:	0.0.0.0

5. Connectivity Verification & Ping Test Logs

End-to-end communication was verified by opening the **Command Prompt** on **PC0 (192.168.10.2)** and issuing ICMP Echo requests to all nodes on the wireless network, including the router gateway and sibling workstations.

```
PC0> ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
PC0> ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
PC0> ping 192.168.10.4
```

```
Pinging 192.168.10.4 with 32 bytes of data:
```

```
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
PC0> ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Reply from 192.168.10.3: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128
```

```
PC0> ping 192.168.10.4
```

```
Pinging 192.168.10.4 with 32 bytes of data:
```

```
Reply from 192.168.10.4: bytes=32 time=2ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
```

```
Ping statistics for 192.168.10.4:
```

```
Packets: Sent = 12, Received = 12, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

--- End of Technical Documentation | Prepared for Network Administration Review ---